

Name: _____

TOPIC TEST

Trigonometric functions (Advanced)

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 10 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

- Convert $\frac{3\pi}{7}$ to degrees.
A 77° **B** 76°
C 0.023° **D** 1.35°
- Find the exact value of $\sin\left(\frac{\pi}{6}\right) + \cos\left(\frac{\pi}{4}\right)$.
A $\frac{\sqrt{2}}{4}$ **B** $\frac{2\sqrt{2}}{3}$
C $\frac{1 + \sqrt{2}}{2}$ **D** $\frac{\sqrt{3} + \sqrt{2}}{2}$
- Find the exact value of $\cos(225^\circ)$.
A $\frac{1}{\sqrt{3}}$ **B** $-\frac{1}{\sqrt{2}}$
C $\frac{1}{\sqrt{2}}$ **D** $-\frac{1}{\sqrt{3}}$
- Find the value of t if $\tan(2t + 15) = \cot(6t - 5)$.
A 90 **B** 20
C 10 **D** 80
- Compared to $y = \cos x$, the function $y = 3 \cos 2x$ has:
A the same amplitude and period
B a larger amplitude and a shorter period
C a smaller amplitude and a shorter period
D a smaller amplitude and a larger period
- A circle has a radius of 25 cm. An arc on the circle measuring 37 cm subtends an angle θ at the centre of the circle. Calculate θ in radians.
A 1.798 **B** 1.480
C 0.676 **D** 84.798

7 Find the exact value of $\sin(-300^\circ)$.

A $\frac{1}{2}$

B $-\frac{1}{2}$

C $\frac{\sqrt{3}}{2}$

D $-\frac{\sqrt{3}}{2}$

8 Given that $\sin \theta = \frac{12}{13}$, find the exact value of $\cos \theta$.

A $\frac{13}{5}$

B not defined

C $\frac{5}{12}$

D $\frac{5}{13}$

9 Which statement is true?

A $\sin\left(x + \frac{\pi}{2}\right) = -\cos x$

B $\cos\left(x - \frac{\pi}{2}\right) = \sin x$

C $\sin\left(x - \frac{\pi}{2}\right) = \cos x$

D $\cos\left(x + \frac{\pi}{2}\right) = \sin 2x$

10 Solve $2\cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$.

A $x = \frac{\pi}{3}, \frac{2\pi}{3}$

B $x = \frac{\pi}{3}$

C $x = \frac{\pi}{6}, \frac{11\pi}{6}$

D $x = \frac{\pi}{6}, \frac{5\pi}{6}$

Part B

10 free-response questions

40 marks

Show your working where appropriate.

- 11** If $\sin \theta = -\frac{3}{7}$, $\cos \theta < 0$ and $\tan \theta > 0$, then find the value of:

a $\cos \theta$

b $\tan \theta$

c θ to the nearest degree

[3 marks]

- 12** Simplify each expression.

a $\frac{\cot x \sec x}{\tan x \cos x}$

b $\frac{\tan^2 x}{\sec^2 x}$

[4 marks]

- 13 a** Prove that $(\sin x - \cos x)^2 = \sec^2 x - \tan^2 x - 2 \sin x \cos x$.

- b** Using what you proved in part **a** or otherwise, prove that $\sec^2 x - \tan^2 x - \operatorname{cosec}^2 x + \cot^2 x = 0$.

[4 marks]

- 14** Prove that $\frac{1 + \sin x}{1 - \sin x} = (\sec x + \tan x)^2$.

[2 marks]

- 15 a** In which quadrant does the angle $\frac{16\pi}{3}$ lie?

- b** Hence find the exact values of $\sin\left(\frac{16\pi}{3}\right)$ and $\cos\left(\frac{16\pi}{3}\right)$.

[4 marks]

- 16** Solve $3 \tan x - 1 = 0$ in the domain $[0, \pi]$, correct to 3 decimal places.

[3 marks]

- 17** Sketch each trigonometric function in the domain $[0, 2\pi]$.

a $f(x) = 3 \cos\left(x - \frac{\pi}{2}\right)$

b $g(x) = \tan x - 1$

c $y = \cos(2x)$

d $f(x) = 1 + \sin 3x$

[8 marks]

18 Find the domain and range of each function.

a $y = -2 \cos x$

b $y = 5 \sin (-3x)$

[4 marks]

19 Find the amplitude and period of $y = -2 \sin 3x$.

[2 marks]

20 Solve the equation $\sin^2 \theta - 4 \sin \theta + 3 = 0$ for the domain $[0, 2\pi]$.

[2 marks]

21 Find the period, amplitude, centre and phase of $y = \sin\left(2x - \frac{\pi}{4}\right) - 3$.

[4 marks]

This is the end of the test.

Use the rest of this page for extra working space.

Answers

- 1** A **2** C **3** B **4** C **5** B
6 B **7** A **8** D **9** A **10** C

11 a $-\frac{2\sqrt{10}}{7}$ b $\frac{3}{2\sqrt{10}}$ c $\theta = 205^\circ$

12 a $\operatorname{cosec}^2 x$ b $\sin^2 x$

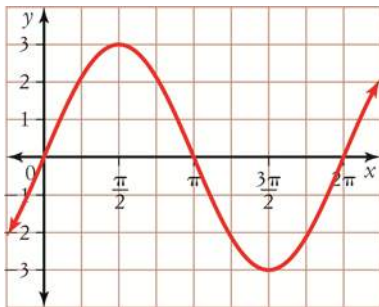
13, 14 Teacher to check.

15 a 3rd quadrant b $-\frac{\sqrt{3}}{2}, \frac{1}{2}$

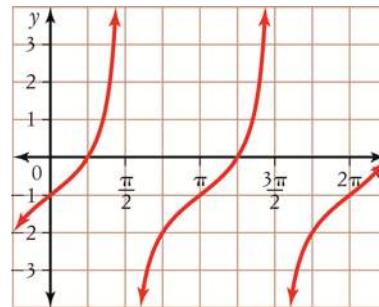
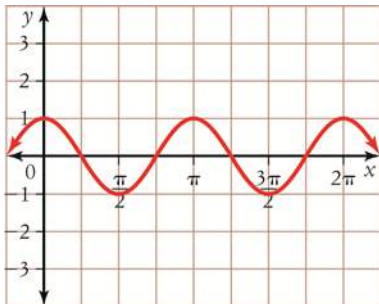
16 $x = 18.435, 198.435$

17 a $f(x) = 3\cos\left(x - \frac{\pi}{2}\right)$

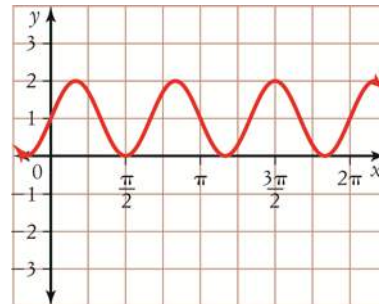
b $g(x) = \tan x - 1$



c $y = \cos(2x)$



d $f(x) = 1 + \sin 3x$



18 a Domain $(-\infty, \infty)$, range $[-1, 1]$

19 Amplitude 2, period $\frac{2\pi}{3}$

20 $\theta = \frac{\pi}{2}$

21 Period π , amplitude 1, centre -3 , phase $-\frac{\pi}{8}$

b Domain $(-\infty, \infty)$, range $[-5, 5]$